Q1 Choose the correct answer.

1 mark for each

1. How many bits are needed to store one BCD digit?

A. 2 bits
B. 4 bits
C. 3 bits
D. 1 bit

2. Convert $(312)_8$ into decimal.

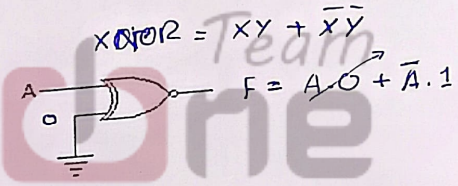
A. $(201)_{10}$
B. $(202)_{10}$
C. $(203)_{10}$
D. $(204)_{10}$

3. What is the addition of the binary number 101001+ 010011=?

A. 010100
B. 111100
C. 000111
D. 101110

4. The following hexadecimal number $(1E,43)_{16}$ is equivalent to

A. $(36.506)_8$
B. $(36.206)_8$
C. $(35.506)_8$
D. $(35.206)_8$



5. For the gate in the given figure the output will be

A. 0
B. 1
C. A'
D. A

6. What is the binary equivalent of the decimal number $(368)_{10}$ => ( )$_2$

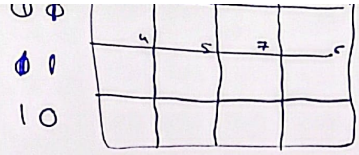
A. 101110000
B. 110110000
C. 111010000
D. 111100000

7. The simplification of the Boolean expression is.

$F = \bar{A}B\bar{C} + A\bar{B}C$

A. A
B. 0
C. 1
D. BC

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$\bar{A}BC\bar{D} = 6$
$0110 =$

8. The Gray code for decimal number 6 is equivalent to

A. 1100
B. 1001
C. 0101
D. 0110

9. The Boolean expression $\bar{A}.B + A.\bar{B} + A.B$ is equivalent to.

$A(\bar{B}+B)$

A. A+B
B. $\bar{A}.B$
C. $\overline{A+B} = \bar{A}+\bar{B}$
D. A.B

$\bar{A}B + A$

$\bar{A}+\bar{B} = \overline{A.B}$

$\overline{Y\bar{Z}} + \overline{X\bar{Y}Z} = \overline{(Y\bar{Z})(X\bar{Y}Z)}$

10. the complement of $f=x'+yz'+xy'z$ is:

$F = \bar{X} + Y\bar{Z} + X\bar{Y}Z$

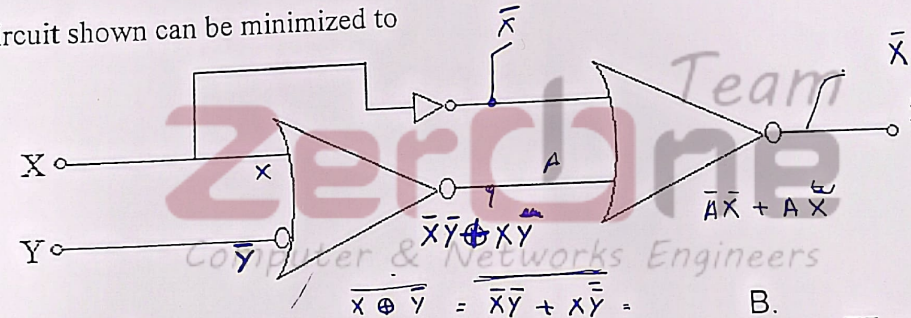
$\bar{F} = X + \overline{Y\bar{Z}} + \overline{X\bar{Y}Z}$

A. x' +(y'+z)(x'+y+z')
B. x'(y+z')(x+y'+z)
C. x(y'+z)(x'+y+z')
D. x+ y'z+x'yz'

$\bar{F} = X + (\bar{Y}+Z)(\bar{X}+Y+\bar{Z})$

$1 + 1 + 1 = 1$

11. The logic circuit shown can be minimized to



$\bar{X}\bar{Y} + XY + \bar{X}$

$\bar{X}(\bar{Y}+1) + XY$

$\bar{X} + XY$

$\overline{X \oplus Y} = \overline{\bar{X}\bar{Y} + X\bar{Y}} =$

$= (\overline{\bar{X}\bar{Y}})(\overline{X\bar{Y}})$

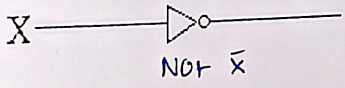
$= (X+Y)(\bar{X}+\bar{Y})$

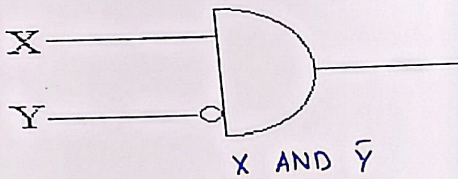
$\bar{A} = X\bar{Y} + \bar{X}Y$

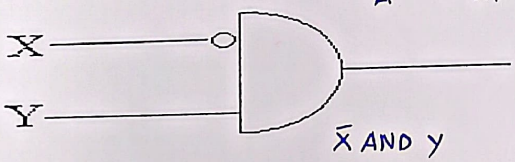
$\bar{X}Y + XY$

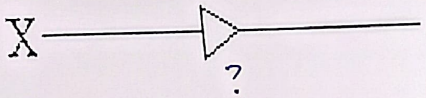
$Y(X+\bar{X})$

Y

A.


B.


C.


D.


12. The output of a logic gate is 1 when all its inputs are at logic 0. the gate is either

A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

A. a NAND or an EX-OR
B. an OR or an EX-NOR
C. an AND or an EX-OR
D. a NOR or an EX-NOR

Fill the answers for Q1 here

1	2	3	4	5	6	7	8	9	10	11	12
B	C	B	C	C	A	C	D	C	C	C	A

Q2) Perform the subtraction on the given numbers using the 2's complement: (3 marks)

$(1110)_2-(10001)_2$

2's comp of (0 0001) → (1 1110)

2's comp = 1 1111

1 1 1 0
+ 1 1 1 1
1 1 1 0 1

1 1 0 1

$= (1110)_2 - (10001)_2$

Q3) Derive the circuits for a three-bit parity generator using an odd parity bit with showing the truth table for the parity generator. (3 marks)

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Q4) Simplify and draw the logic diagram for the given expression. (4 marks)

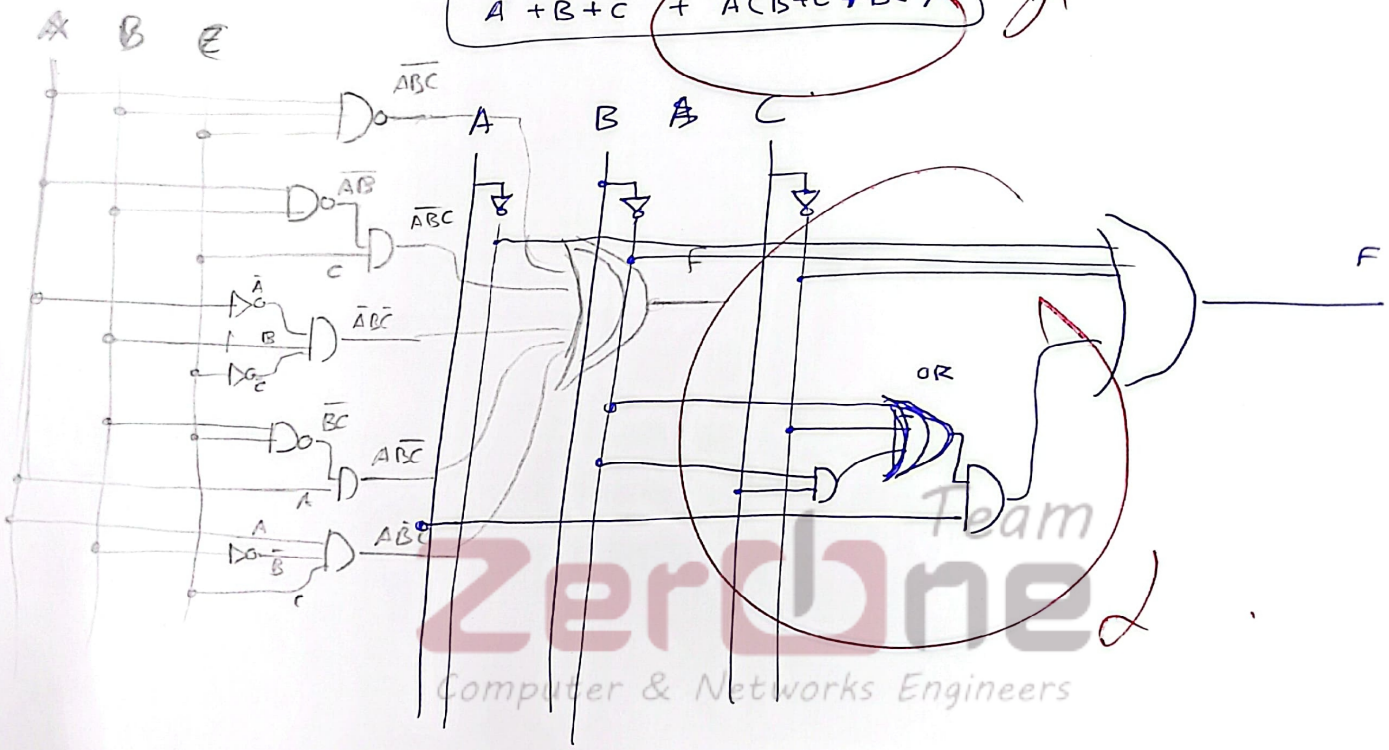
$F = \overline{ABC} + \overline{AB}C + \bar{A}B\bar{C} + A\overline{BC} + A\bar{B}C.$

$F = \bar{A} + \bar{B} + \bar{C} + (\bar{A} + \bar{B})C + \bar{A}B\bar{C} + AC(\bar{B} + \bar{C}) + A\bar{B}C$

$\bar{A} + \bar{B} + \bar{C} + \bar{A}C + \bar{B}C + \bar{A}B\bar{C} + A\bar{B} + AC + A\bar{B}C$

$\bar{A}(1+C) + \bar{B}(1+C) + \bar{C}(1+\bar{A}B) + A\bar{B} + A\bar{C} + A\bar{B}C$

$\bar{A} + \bar{B} + \bar{C} + AC(\bar{B} + \bar{C} + B\bar{C})$



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(c) $F = A\bar{B} + C + A\bar{C}\bar{D}$

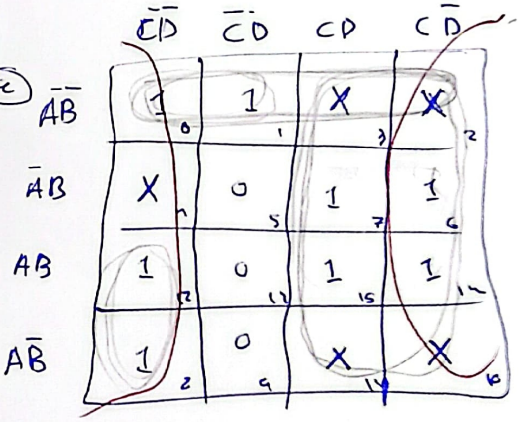
Q5) For the given Boolean function (8 marks)

$F(A,B,C,D) = \bar{A}\bar{B}\bar{C} + A\bar{C}\bar{D} + BC$

000 100 $\bar{A}BC$ + ABC (011 111)

Which has the don't care condition

$d(A,B,C,D) = \sum(2,3,4,10,11)$



a. Draw the K-map for function f (2marks)

b. Minimize f as product-of-sum (without don't care) (2marks) $F = \prod M = (5, 9, 12) = \bar{A}B\bar{C}D +$

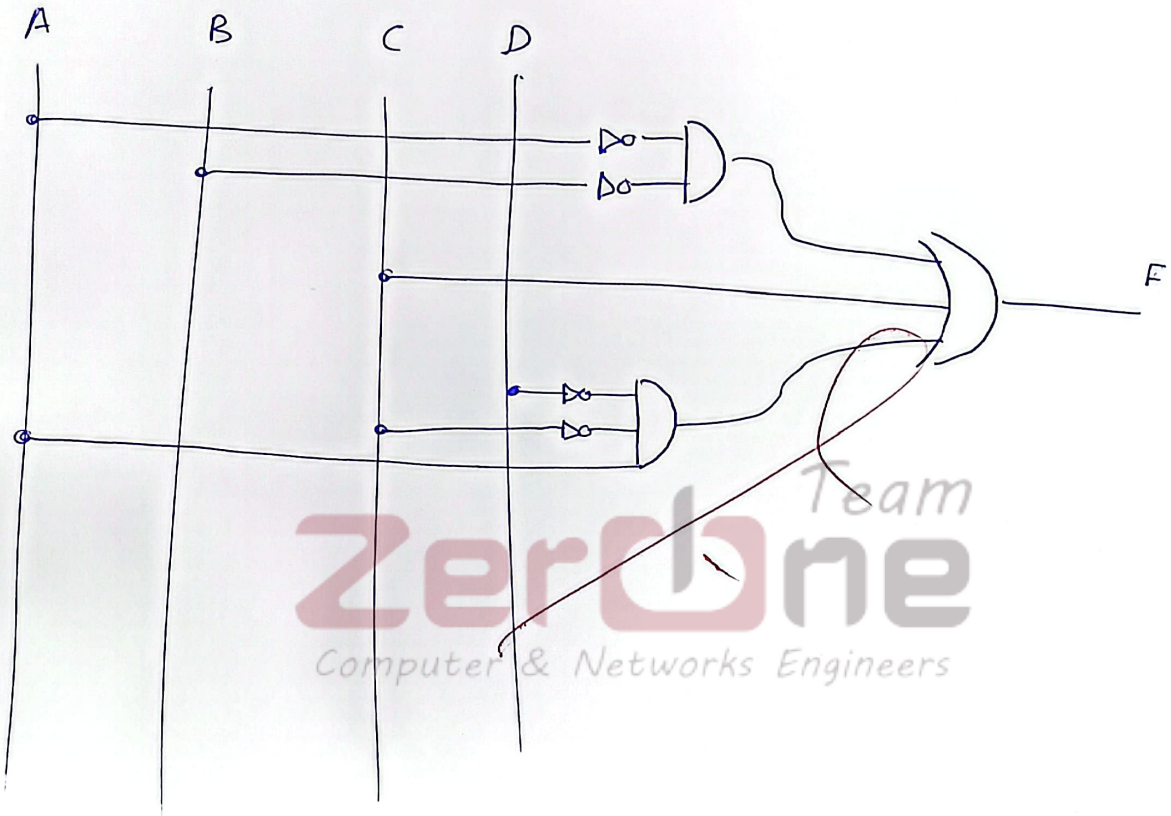
c. Minimize f as sum-of-products (with don't care) (2marks) $F = \sum m =$

$F = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}BC\bar{D} + \bar{A}BCD + ABC\bar{D} + ABCD$

m_0 m_1 m_8 m_{12} m_6 m_7 m_{14} m_{15}

d. convert and draw the minimized f in branch (c) to NAND gate only circuit (2 marks)

$$F = \bar{A}\bar{B} + C + A\bar{C}\bar{D}$$



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Q1: 1) 1 BCD digit → 4 bits. B.

2) $(312)_8 = 2\times1 + 1\times8 + 3\times64$ B.
$= 10 + 192 = (202)_{10}$.

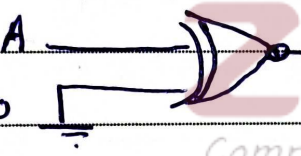
64 × 3 = 192

3)
$$\begin{array}{r} 101001 \\ 010011 + \\ \hline 111100 \end{array}$$
B.

4) $(1E.43)_{16}$?

$(0001\ 1110.0100\ 0011)_2$

$(0\ 3\ 6.2\ 0\ 6)_8$. B.

5) 

C.

$x = A0 + A'0' = 0 + A'1 = A'$

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6) $(368)_{10}$ → binary

368	0
184	0
92	0
46	0
23	1
11	1
5	1
2	0
1	1
0	

$(101110000)_2$ A.

7) $F = \overline{A'BC'} + \overline{AB'C} = A+B'+C+A'+B+C'$

$= (A+A') + (B+B') + (C+C') = 1+1+1 = 1$. C.

8) $(6)_{10} \longrightarrow (0110)_2 \longrightarrow (0\ 1\ 0\ 1)_{gc}$.

XOR XR XR

from binary to BCD — MSB — نفس

$0 \oplus 1 \quad 1 \oplus 1 \quad 1 \oplus 0$

C.

9) $A'B + ACB' + B) = A + A'B = (A+A')(A+B)$. A.

10) $\bar{f} = \overline{x' + yz' + xy'z} = x(\overline{yz'})(\overline{xy'z})$

$= x(y'+z)(x'+y+z')$. C.

11) $\overline{\left(\overline{(x+y')} + x'\right)} = x(x+y') = x+xy' = x(1+y') = x$

D. $x \longrightarrow$ buffer (same output for the input)

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12) output is 1 $\longrightarrow$ all inputs are 0

A	B	F
A' 0	B' 0	1 ← using ~~POS~~ SOP
0	1	0
1	0	0
1	1	0

$F = A'B' = \overline{A+B}$

inverters + AND — NOR

D.

Q2: $0\ 1\ 1\ 1\ 0$ 14

$0\ 1\ 1\ 0\ 1$ + -17

$1\ 0\ 1\ 0\ 1$

convert back $\hookrightarrow -(00011)_2 \longrightarrow (-3)_{10}$

Q4: $F = A' + B' + C' + (A' + B')C + A'B'C'$

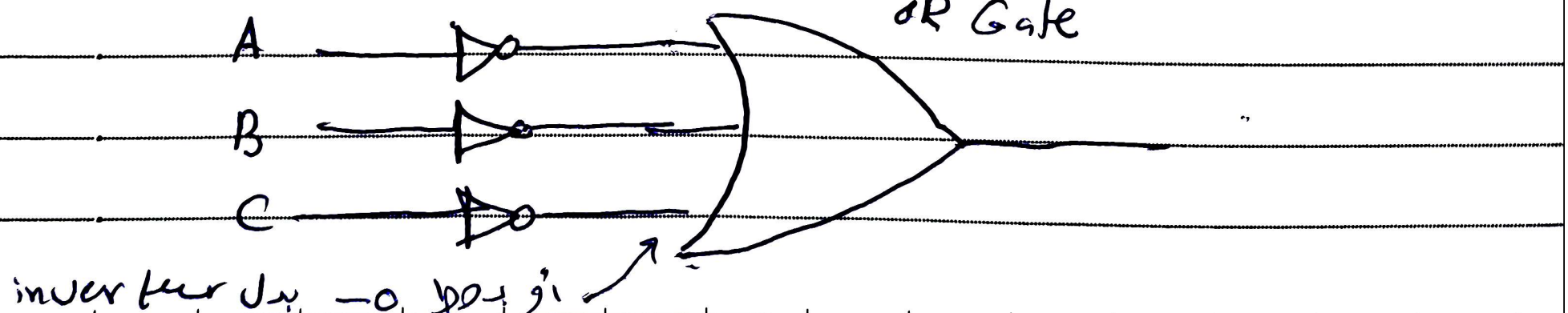
$$+ A(B' + C') + AB'C$$

$$= A' + B' + C' + A'B'C' + AB' + AC' + A'C + B'C + AB'C$$

$$= C'(A'B' + A + 1) + A' + B'(1 + A + C + AC) + A'C$$

$$= C' + A'(1 + C) + B' = A' + B' + C'$$

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Q5: $F = A'B'C' + AC'D' + BC$

$d = \sum (2, 3, 4, 10, 11)$

$= A'B'C'(D+D') + AC'D'(B+B')$

$+ BC(A+A')(D+D')$

1 0 12 8

$= A'B'C'D + A'B'C'D' + AB'C'D' + AB'C'D'$

0 0 0 1 0 0 0 0 1 1 0 0 1 0 0 0

$+ ABCD + ABCD' + A'BCD + A'BCD'$

1 1 1 1 1 1 1 0 0 1 1 1 0 1 1 0

15 14 7 6

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$F = \sum m(0, 1, 6, 7, 8, 14, 15)$.

$d = \sum m(2, 3, 4, 10, 11)$. ← don't care.

a)

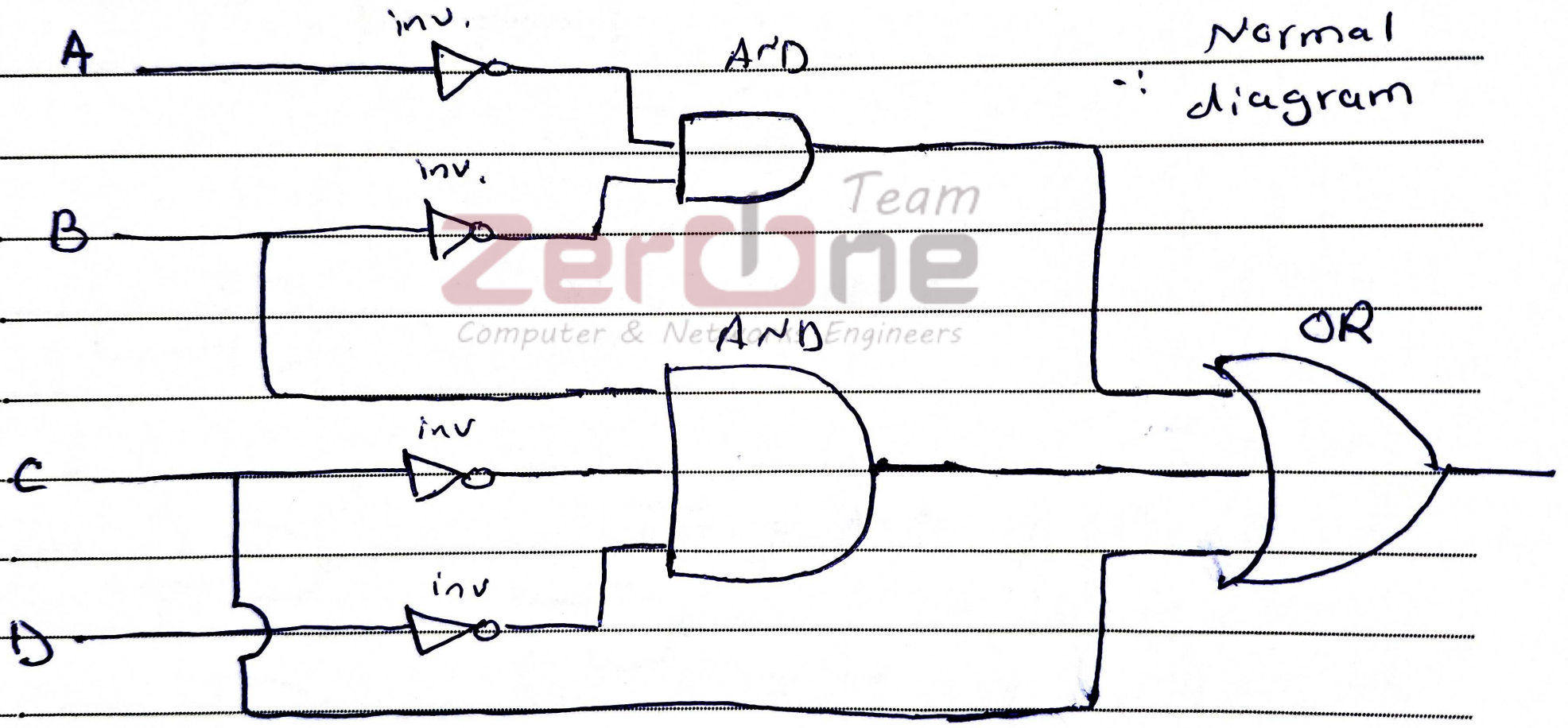
AB \ CD	C'D' 00	01	C 11	C 10
A'B' 00	1 m_0	1 m_1	X m_3	X m_2
01 B	X m_4	0 m_5	1 m_7	1 m_6
11 B	1 m_8	0 m_9	X m_{11}	X m_{10}
10	0 m_{12}	0 m_{13}	1 m_{15}	1 m_{14}

b) $F = (B' + CD')(C + A'B)$ pos w/out x

c) $F = C + BC'D' + A'B'$ SOP with x

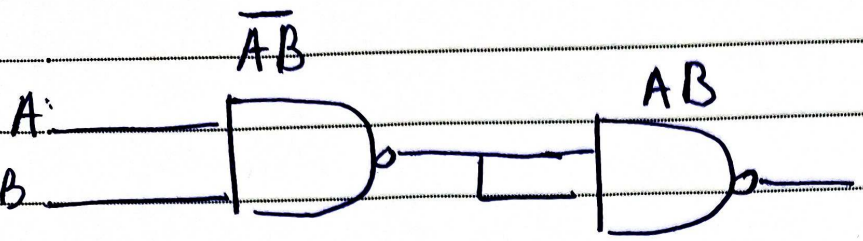
d) branch c: $F = C + BC'D' + A'B'$

draw using only NAND Gates.



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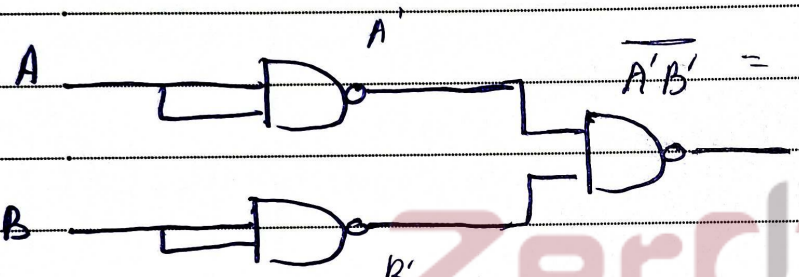
↝ Remember :



AND using NAND



inverter using NAND



OR using NAND (same for 3 inputs).

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